

Obstacle Negotiation & **Rule Compliance Analysis**

Objective



Analyze recurring failure patterns impacting perception and rule compliance.

Dataset Summary

Number of Cases Observed :100+

Examples stored:13 presently

Scenario-focus: Intersection and Obstacle Negotiation

Key Issue Areas:

1.Perception under lighting conditions

2.Obstacle negotiations at the intersections.

3.Intent related errors

(Additional patterns under observation, pending further validation)

Key Patterns

★ 1.Decision failure

Interaction that happen near the intersections lead to Poor rule compliance such as violation of stop sign/ traffic lights.

★ 2.Perception failure/ Low perception confidence

Struggles reading the traffic lights and VRUs in bright environments, such as camera glare, Sun glare, Shadows.

★ 3.Timing/Prediction Issues

Delayed perception or early action before the intent is clear.

Pre-Intersection Negotiation Leads to Rule Compliance Issues

- Observed during interactions with lead vehicles and parked vehicles(active/inactive) near intersections
- Focus shifts to surrounding vehicle behavior then to the traffic control categories
- Leads to delayed or incorrect stopping decisions
- Results in traffic signal or stop sign violations

Likely Causes of Observed Failures

- Competing priorities between vehicle interaction and traffic rule enforcement
- Perception uncertainty in complex or low-visibility conditions (e.g., glare, shadows)
- Delayed or incomplete interpretation of surrounding agent intent

Impact of Observed Failures

- Leads to traffic rule violations at intersections where the area is often highly populated (signals / stop signs).
- Brings-in unsafe or undesirable behavior in high-risk scenarios.
- Reduces reliability in complex interaction environments (eg;narrow hilly routes).
- May impact overall system trust and predictability in real-world deployment.

Suggested Improvements

- Broaden the training data to properly cover those complex, high-interaction intersection scenes.
- Fix the labeling guidelines so we're clear on how to negotiate obstacles near signals.
- Better handle visibility issues like heavy glare and shadows that affects the sensor perception.
- Improve how we read the intent of other drivers during tight, close-range moves.

Case Links for Reference

Confidential

Additional links